



ME 207: Fluid Dynamics

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Lab Experiment 4: The Bernoulli's Apparatus

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Group 14

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Abstract

This experiment aims to verify the Bernoulli equation and report the deviations between the experimental values and those predicted by Bernoulli's equation. The modifications required to be made in Bernoulli's equation are also elaborated.

1 Task 4.1 (Essential Background)

Q1: Read up on Bernoulli's equation in Fluid Mechanics. Fundamentally, what is Bernoulli's equation about? What are the assumptions under which this equation is valid?

Ans. Bernoulli's equation in fluid dynamics is a conservation of energy in a fluid in motion. It equates pressure, velocity, and Height (or potential energy) in a fluid on a stream. In other words, it describes how fluid energy is shared among kinetic energy, potential energy, and pressure energy. The Bernoulli's equation is:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \quad (1)$$

Where:

- P = Pressure energy per unit volume (Pa)
- $\frac{\rho v^2}{2}$ = Kinetic energy per unit volume (J/m³)
- ρgh = Potential energy per unit volume due to height (J/m³)
- ρ = Fluid density (kg/m³)
- v = Flow velocity (m/s)
- g = Acceleration due to gravity (m/s²)
- h = Height above a reference level (m)

This equation tells us that the total mechanical energy of an incompressible, non-viscous fluid remains constant along any streamline.

Below are the assumptions that are made to make this equation valid:

- Incompressible Flow: Fluid density is constant.
- Non-viscous (Ideal) Fluid: the fluid has no internal friction or viscosity.
- Steady Flow: The fluid's properties at any given point remain constant over time.
- No Energy Loss: No energy can be added from external sources

Q2: To circumvent the assumptions, several modifications to Bernoulli's equation have been proposed. Read up on these modifications from your Fluid Mechanics textbook and report them in your submission

Ans. Several modifications must be made to study the flow using Bernoulli's equations. Here are the changes to Bernoulli's equation that help to get accurate results in our experiment:

1. **Energy Loss due to Friction:** In the non-ideal fluid, there is some energy loss due to the viscosity. This can be addressed by adding the new term called head loss(h_f).

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2 + h_f = 0 \quad (2)$$

2. **Unsteady Flow:** For the flows that vary with time, a time-dependent term called the velocity potential function has to be added. It is denoted by ϕ .

$$\frac{\partial \phi}{\partial t} + \frac{P}{\rho} + \frac{v^2}{2} + gh = \text{constant} \quad (3)$$

3. **External Work:** When flow is due to mechanical devices like pumps or turbines, they can add or subtract the energy. So the work(W : work per unit volume) term can be added.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 + W = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2 = 0 \quad (4)$$

Q3: What is head loss in a pipe flow? Why is it important? Can you quantify head loss?

Head loss in a pipe flow is basically a loss of energy in a fluid because of friction between the fluid and the walls of the pipe. Energy loss also happens due to some minor reasons like fittings, bends, valves, or other obstructions. In short, when a fluid flows through a pipe, it loses some energy due to resistance which reduces its pressure and flow efficiency.

Head loss is important factor in the experiment to maintain the system efficiency. Excessive head loss increases the energy required to maintain flow, which can be result in higher pumping costs. Also, pipe is selected based on the value of the head loss. It also predicts where the pressure will drop which can prevent system failures.

There are two types of head loss: Major head loss and Minor head loss.

1. **Major head loss:** It occurs due to the friction between the fluid and the pipe wall. It can be calculated by Darcy's equation. This equation determines the losses in the pipe.

$$H_f = f \frac{L}{D} \frac{v^2}{2g} \quad (5)$$

- H_f = Head loss (m)
- f = Darcy friction factor (dimensionless)
- L = Length of the pipe (m)

- D = Diameter of the pipe (m)
- v = Flow velocity (m/s)
- g = Acceleration due to gravity (9.81 m/s^2)

2. **Minor head loss:** This type of head loss occurs due to valves, bends, expansions and fittings in the pipe system. It can be calculated using this equation:

$$h_m = K \frac{v^2}{2g} \quad (6)$$

Where :

- H_m = Minor head loss (m)
- K = Loss coefficient (dimensionless), specific to each fitting
- v = Flow velocity (m/s)

The total head loss can be calculated by the sum of both major and minor losses:

$$H_{total} = H_f + \Sigma H_m \quad (7)$$

Q4: What is friction factor and what does it depend on?

The friction factor (f) is a dimensionless parameter characterizing the friction (or opposition) fluid experiences while flowing in a pipe. It is crucial in the measurement of head loss due to friction in pipes, especially in the Darcy-Weisbach equation, (which is shown above as well):

$$H_f = f \frac{L}{D} \frac{v^2}{2g} \quad (8)$$

- The friction factor depends on the following factors: Reynolds Number (Re): It determines whether the flow is laminar or turbulent.
- Relative Roughness ($\frac{\epsilon}{D}$) (if the flow is Turbulent in the rough pipe): It is the ratio of the pipe's surface (ϵ) to its diameter (D).

Q5: How do you evaluate the friction factor for a given flow? Give examples.

The friction factor can be determined by following the steps given below:

- Step 1: Calculate the Reynolds Number (Re)
- Step 2: Determine the type of flow and use the Appropriate Formula to find friction factor(f)

The friction factor for different types of flow rates is given by :

For the laminar flow :

$$f = 64/Re$$

Where:

- f = friction factor
- Re = Reynolds number

For the turbulent flow (smooth pipe) :

$$f = 0.0791 Re^{-0.25}$$

Above result comes from the Blasius equation.
Where:

- f = friction factor
- Re = Reynolds number

For the turbulent flow (rough pipe) :

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left[\frac{e}{3.7D} + \frac{2.51}{Re\sqrt{f}} \right]$$

The above result comes from the Colebrook equation. Where:

- f = friction factor,
- Re = Reynolds number,
- e = relative roughness($\frac{D}{\epsilon}$),
- D = Diameter of the pipe.

2 Introduction



Figure 1: Image of the experimental setup

Bernoulli's experiment is designed to study the difference in pressure and velocity of a fluid as it flows through a converging-diverging nozzle. Bernoulli's equation is a fundamental concept in fluid dynamics. The principle is named after the mathematician/Scientist Daniel Bernoulli.

Bernoulli's principle states that the sum of all forms of energy (pressure head + velocity head + elevation head) of a fluid is the same at any two points along a streamline for an ideal, incompressible and inviscid fluid and if the flow is steady

These assumptions cause errors in the experimental and theoretical results. This equation is derived using the conservation of energy principle. If the velocity of the fluid increases, then simultaneously, the pressure decreases, or the potential energy decreases. We can compare the theoretical and experimental results by measuring pressure at different points along the pipe using manometers. Causes of error are real-world factors such as head loss, viscous effects, turbulence, etc. Bernoulli's equation is used to find the fluid's velocity, pressure difference, flow rate, height difference, etc. Pitot's tube, orifice flow meter, nozzle flow meter, and venturi flow meter use Bernoulli's equation to determine the flow rates. This experiment follows Bernoulli's principle.

Below attached is the schematic diagram of the apparatus.

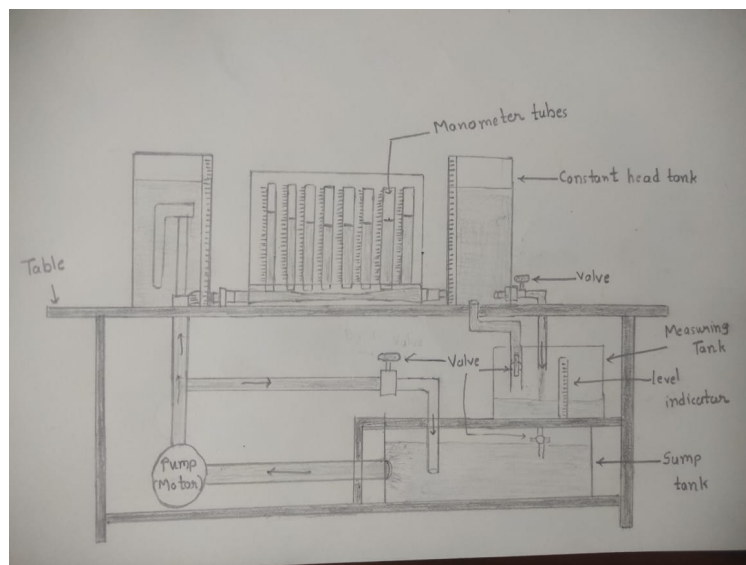


Figure 2: Schematic diagram of the apparatus

3 Objectives

- (i) Study the pressure variations in flow through a converging diverging nozzle.
- (ii) Compare the observations with Bernoulli's equation.
- (iii) Interpret the results and the potential discrepancies between theoretical analysis and observed data.

4 Experimental Procedure

1. Firstly we have to fill in enough water in the bottom tank so that the flow of water is consistent throughout the apparatus; This step is essential because if less water is present, then the pump will start sucking in the air due to lack of water leading to inconsistent flow rate.
2. Secondly, we control the flow rates using two valves, which will help adjust the water height in the two head tanks
3. It is essential to ensure that the water height in the head tank is constant after we calibrate the flow rates. To double-check this step, we wait 2-3 minutes to check for any differences in height observed; if the height changes, we have to adjust the flow rates accordingly.
4. Once the flow rate is constant, we measure the height of the subsequent tubes in the horizontal tube.
5. The area of the cross-section decreases initially, reaches a minimum value and then starts to increase to the initial value of the area; according to the continuity principle, the velocity of water should increase as the area decreases. As the velocity increases, the pressure should decrease according to Bernoulli's principle, and this pressure drop is indicated by the decrease in height in the tubes.
6. As the tubes are not perfectly on the same level, we have to add a correction factor to some of them.
7. The next step is to calculate the flow rate of the water through the apparatus, for which we will calculate the time the water takes to fill a particular height in the sink tank in which a narrow transparent tube is attached with markings.
8. We must close the sink's bottom valve so the water does not leak through the tank.
9. The time water takes to fill a particular height is measured, and multiple readings are to be taken to minimize the error.
10. As we know the area of the sink tank and the time it takes to fill a particular height, we can calculate the flow rate.
11. The same experiment will be repeated for a different flow rate, controlled by adjusting the valve.

5 Results and Error Analysis

5.1 Results

5.1.1 For the first flow rate

The values of heads in the left and right tanks are given below.

- $H_L = 330$ mm
- $H_R = 314$ mm
- $\Delta H = H_L - H_R = 16$ mm

Calculation of the flow rate

The flow rate is calculated using the volume occupied by water in a given time interval. The time required for water in the measuring tube attached to the measuring tank to be filled by 0.5 cm is measured, and then the flow rate is measured using the formula given below.

$$Q = \frac{lbh}{t}$$

Where:

- l = length of the measuring tank (m).
- b = breadth of the measuring tank (m).
- h = elevation of water in the measuring tube (m).
- t = time required to achieve the elevation(m).

For the first flow rate, the measured time values are as follows.

Time(s)	Elevation (cm)
40.71	0.5
40.19	0.5
50.4	0.5
52.17	0.5

Thus, the value of time turned out to be

$$t = 45.87s$$

Also, the length and the breadth of the measuring tank are given, which are

- $l = 0.4$ m
- $b = 0.3$ m

Hence, the flow rate is given by

$$Q_e = 0.00013 \frac{m^3}{s}$$

We can use this value of flow rate to calculate the velocity of the flow in the pipe, which can be further used to calculate the velocity head. Thus, the velocity head can be written in terms of flow rate, which is demonstrated below.

$$K = \frac{v^2}{2g} = \frac{Q^2}{2gA^2}$$

- K = Kinetic head (m)
- v = Flow velocity (m/s)
- Q = Flow rate (m^3/s)
- A = Area of the cross section (m^2)

Also, the pressure head is given by

$$V = \frac{P}{\rho g}$$

Where:

- P = Pressure (Pa).
- ρ = Density of the fluid (Kg/m^3)

We consider the reference plane such that the axis of the pipe passes through the reference plane, and thus the potential head is zero across the pipe since the pipe is horizontal.

Thus, the total head is given by

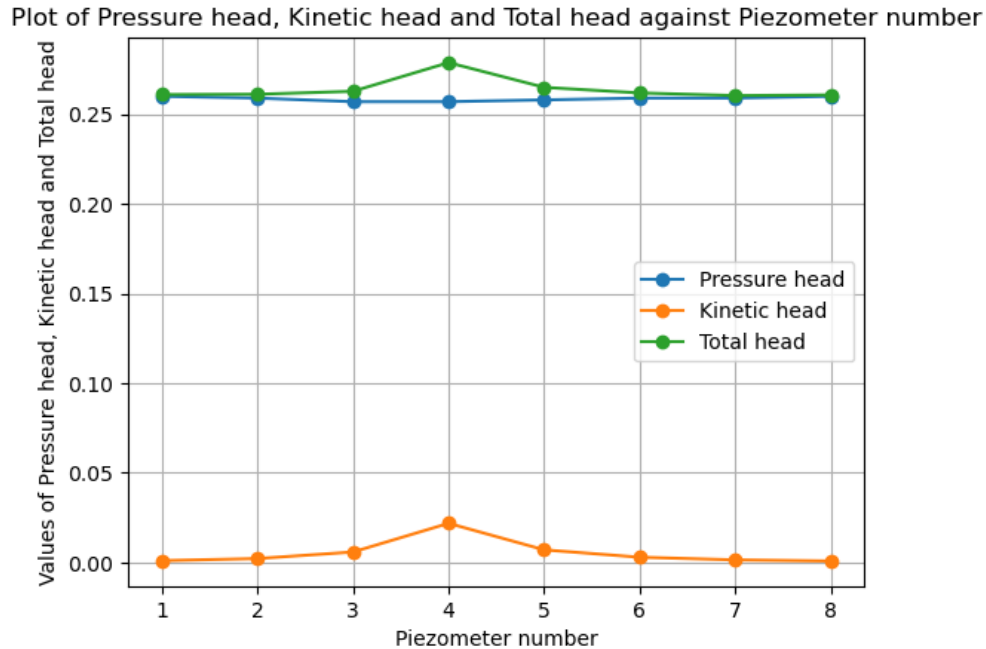
$$T = K + V$$

Below are the values of the Pressure head, Kinetic head, and Total head for the different sections of the pipe.

Piezometer	Area (m^2)	Height(m)	Velocity(m/s)	K	V	T	σ_T
1	0.00095	0.26	0.14	0.001	0.26	0.26	0.0021
2	0.00068	0.259	0.2	0.002	0.259	0.261	0.0011
3	0.000388	0.257	0.34	0.006	0.257	0.263	0.0009
4	0.0002	0.257	0.65	0.02	0.257	0.28	0.0179
5	0.000353	0.258	0.37	0.007	0.258	0.265	0.0029
6	0.00055	0.259	0.24	0.003	0.259	0.262	0.001
7	0.000789	0.259	0.16	0.001	0.259	0.26	0.0021
8	0.00178	0.26	0.12	0.0007	0.26	0.26	0.0021

K, V, and T are Kinetic head, Pressure head, and Total Head, respectively, and they are measured in meters.

From the table shown above, the values of the Pressure head, Velocity head, and total head can be plotted against each section where the piezometer is connected. The graph is shown below.



5.1.2 For the second flow rate

The values of heads in the left and right tanks are given below.

- $H_L = 241 \text{ mm}$
- $H_R = 206 \text{ mm}$
- $\Delta H = H_L - H_R = 35 \text{ mm}$

Calculation of the flow rate

The flow rate is calculated using the volume occupied by water in a given time interval. The time required for water in the measuring tube attached to the measuring tank to be filled by 0.5 cm is measured, and then the flow rate is measured using the formula given below.

$$Q = \frac{lbh}{t}$$

Where:

- l = length of the measuring tank (m).
- b = breadth of the measuring tank (m).

- h = elevation of water in the measuring tube (m).
- t = time required to achieve the elevation(m).

For the first flow rate, the measured time values are as follows.

Time (s)	Elevation (cm)
22.60	0.5
17.34	0.5
20.65	0.5
22.21	0.5

Thus, the value of time turned out to be

$$t = 20.70s$$

Also, the length and the breadth of the measuring tank are given, which are

- $l = 0.4$ m
- $b = 0.3$ m

Hence, the flow rate is given by

$$Q_e = 0.00029 \frac{m^3}{s}$$

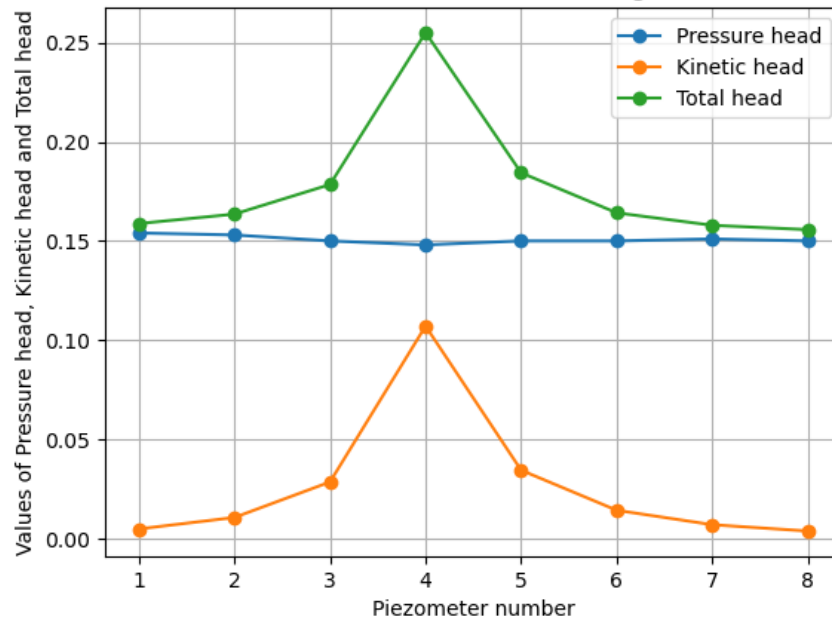
Using this value, the pressure head, the kinetic head, and the total head can be calculated using a similar approach to the first flow rate.

Piezometer	Area (m^2)	Height(m)	Velocity(m/s)	K	V	T	σ_T
1	0.00095	0.154	0.31	0.005	0.154	0.159	0.0229
2	0.00068	0.153	0.45	0.01	0.153	0.160	0.0219
3	0.000388	0.150	0.75	0.03	0.15	0.18	0.0019
4	0.0002	0.148	1.45	0.1	0.148	0.2	0.0181
5	0.000353	0.15	0.82	0.03	0.15	0.18	0.0019
6	0.00055	0.150	0.53	0.01	0.15	0.160	0.0219
7	0.000789	0.151	0.37	0.007	0.151	0.158	0.0239
8	0.00178	0.152	0.27	0.004	0.152	0.156	0.0259

K, V, and T are Kinetic head, Pressure head, and Total Head, respectively, and they are measured in meters.

Also, the figure showing the plots of the Pressure head, Kinetic head, and Total head is shown below.

Plot of Pressure head, Kinetic head and Total head against Piezometer number



5.2 Calculation of errors

The upper bound estimate was used to calculate the error.

It is given by the expression shown below:

$$\sigma_Y = \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i} \right| \sigma_i$$

Here, Y is the function of independent variables $x_1, x_2, \dots, x_n$.

The above formula is used to calculate the least count errors of all the entities, using the least count of the below entities known.

Entity	Least count error
h (elevation in measuring tank)	0.001 m
t (time taken by water to rise by h)	0.01 s

The least-count errors in the other entities are shown in the table below.

Entity	Least count error
Q (Flow rate)	$0.00003 m^3/s$
v (Flow velocity)	$0.00003 m/s$
K (Kinetic head)	$0.00001 m$
T (Total head)	$0.01001 m$

Now, the total error in the calculation of the total head can be expressed as the sum of the average error and the least count error.

$$\sigma_T = \sigma_{avg} + \sigma_{LC}$$

Now, the total error in the calculation of the total head can be expressed as the sum of the average error and the least count error.

$$\sigma_{avg} = \frac{1}{n} \sum_{i=1}^n |T_{avg} - T|$$

The percentage error in the Total head can be calculated for both the flow rates using the expression shown below.

$$\% \sigma_T = \frac{\sigma_T}{T} * 100$$

	First flow rate	Second flow rate
Average value of total Head	0.2639 m	0.1691 m
Error in the measured value	0.0037 m	0.0173 m
Least count	0.01002 m	0.01001 m
Absolute Error in Head	0.01371 m	0.03741 m
Percentage Error	5.20 %	22.12 %

Table 1: A descriptive caption for your table

5.3 Comparison of theoretical and analytical results

The Bernoulli's head equation is given as:

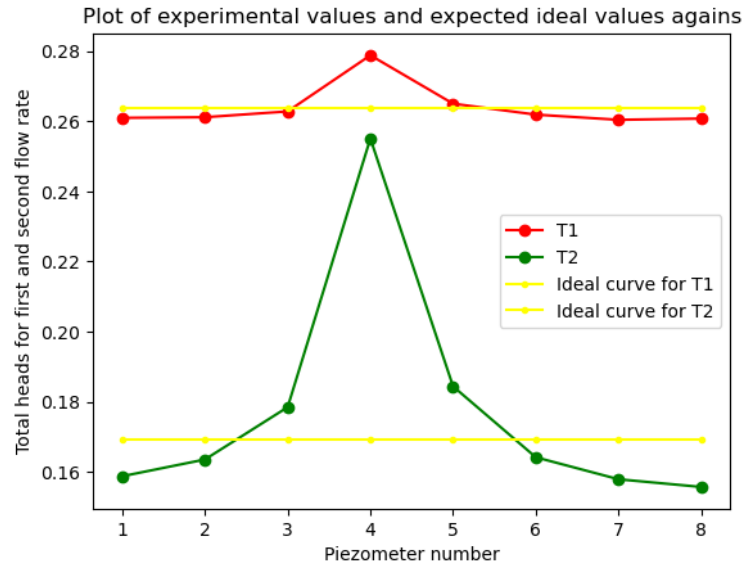
$$\frac{v^2}{2g} + \frac{p}{\rho g} + z = \text{constant}$$

Since the reference frame is chosen so that the axis of the pipe passes through the reference,

$$\frac{v^2}{2g} + \frac{p}{\rho g} = \text{constant}$$

Thus, according to the theory, the summation of the Pressure head and Kinetic head should remain constant.

However, it was experimentally observed that there were slight deviations from this behavior. The graph indicating the expected curve and experimentally observed curve is given below.



In order to account for this, several factors like head loss, external work, and unsteady flow can be considered, as mentioned in the first section of the report. In this experiment, values like friction factor and Reynolds number are not known. Thus, head loss cannot be quantified.

5.4 Causes of error

1. **Leakage in measuring tube** - The tube connecting the measuring tube with the measuring tank was loose, causing some water to leak from the tube; this could lead to uncertainty in measuring the flow rate experimentally.
2. **Viscous losses** - As fluid flows through the pipe, some Kinetic energy is lost because of the interaction between fluid and the pipe walls. These losses are accounted for in viscous losses.
3. **Losses due to valves and bends** - Components like joints and valves can cause turbulence in the flow and lead to a loss of kinetic energy; such losses are hard to determine and cause a mismatch in the results.
4. **Pump was sucking air** - In some instances, when the water level dropped below a certain level, the pump present near the sump tank started absorbing air; this could lead to a change in head levels, causing disturbance to the flow. This contradicts the assumption of steady flow when using Bernoulli's equation.
5. **Imperfections in fluid** - The fluid used to experiment was water. However, some impurities, such as dirt and air bubbles in the fluid, could contribute to the total error. Whereas Bernoulli's equation is valid for an ideal incompressible fluid.
6. **Least count errors** - The number called least count is associated with every measuring instrument; errors arising from the least count are termed as least count errors. The

contribution of the least count errors to the total error depends upon the instrument's resolution.

7. **Response error** - When we measured the time for the specific height reached in the tank to determine the flow rate, the time was measured manually with the help of a watch. So, it can lead to a delay in time measurement.
8. **Instrumental inaccuracies** - There were minor defects in the experimental apparatus, such as pipes being bent and valves being loose, which could eventually cause uncertainties in the results.
9. **Air trapped in tubes** - While an attempt was made to maintain a steady flow across the pipe, tiny air bubbles appeared in some instances. Some of these could have gotten trapped inside the tubes, which could cause uncertainties in measuring pressure.
10. **Parallax error** - This error is caused when the observer takes measurements from an angle, resulting in differences from the actual reading.

6 Observation

It was observed that the Bernoulli equation was approximately satisfied at the sections corresponding to the corners of the pipe. However, significant deviations were observed at the center, where the area of the pipe changed drastically.

7 Conclusion

After observing the values of the total head of the water flowing through the pipe at different cross sections, it was concluded that Bernoulli's equation serves as a good approximation that relates velocity with the pressure. However, several modifications, like the inclusion of major losses and minor losses, can be considered to circumvent the assumptions of Bernoulli's equation.

8 Acknowledgements

We want to thank our Teaching Assistant, Mr. Manoranjan, for explaining the concept of Bernoulli's equation and providing us with essential insights about sources of errors.

References

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